

GENX: Porewater Ammonium

May 2026 Samples

2026-07-23

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##Run Information

```
cat("Run Information: Brenden Blakley") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
library(here)
```

here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "May"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
NH3_files <- c("Raw Data/20260608_2m_GENXrr_May_NH4.csv",  
              "Raw Data/20260624_GENXrr_2Mrr_May2026.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
file_path <- here("Porewater QAQC Logs", "GCREW_NH4_QAQC_Log.csv")
```

```
file_path_QAQC <- here("Porewater QAQC Logs", "GCREW_SEAL_Check_QAQC_Log.csv")
```

```
final_path <- "Processed Data/GCREW_GENX_Porewater_NH4_202605.csv"
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- "20260608 run stopped part way through; ch1-80 not collected"
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

20260608 run stopped part way through; ch1-80 not collected

##Setup

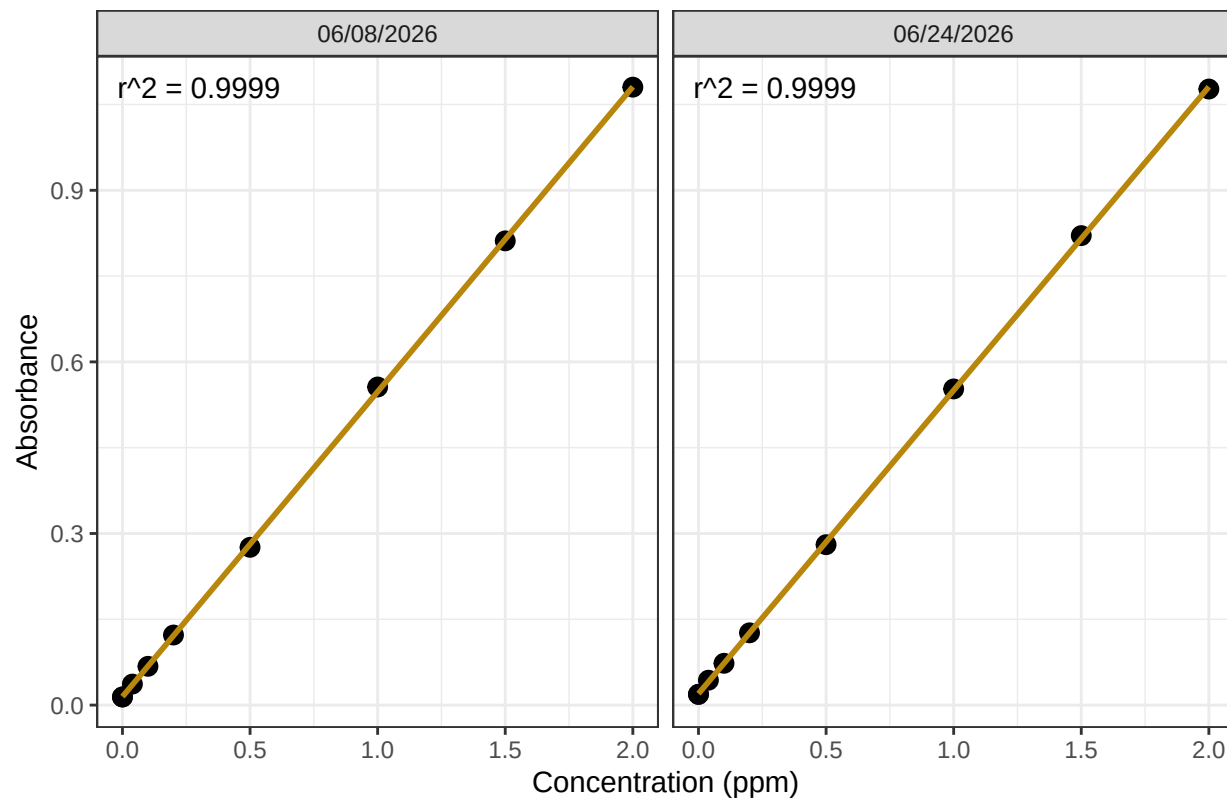
##Read in metadata and create similar sample IDs for matching to samples

```
##Import Data & Clean
```

0.1 Assessing standard Curves

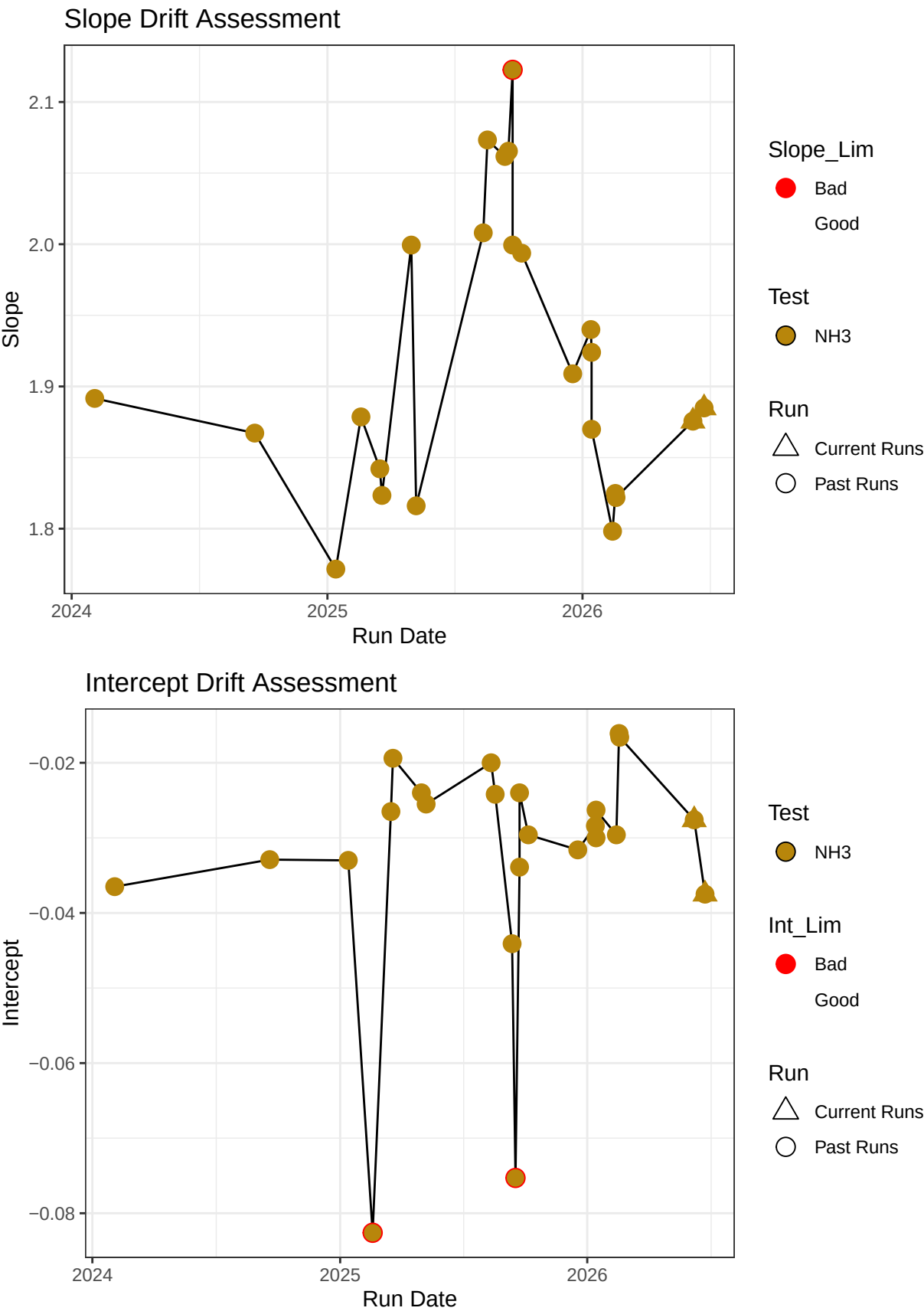
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NH3 Standard Curve



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	06/08/2026	R2_Pass - PROCEED	0.9998997	1.875603
Ammonia 2	06/24/2026	R2_Pass - PROCEED	0.9999260	1.884907

0.2 Update Standard Log



0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: c(\"Ch4-80\", \"Ch5-80\", \"Ch6-40\", \"Ch6-80\", \"Ch8-40\", \"Ch8-80\", \"Ch9-
```

```
## [1] "Diluted Samples: c(\"Ch4-80\", \"Ch5-80\", \"Ch6-40\", \"Ch6-80\", \"Ch8-40\", \"Ch8-80\", \"Ch9-40\", \"Ch9-80\", \"Ch10-40\", \"Ch10-80\")"
```

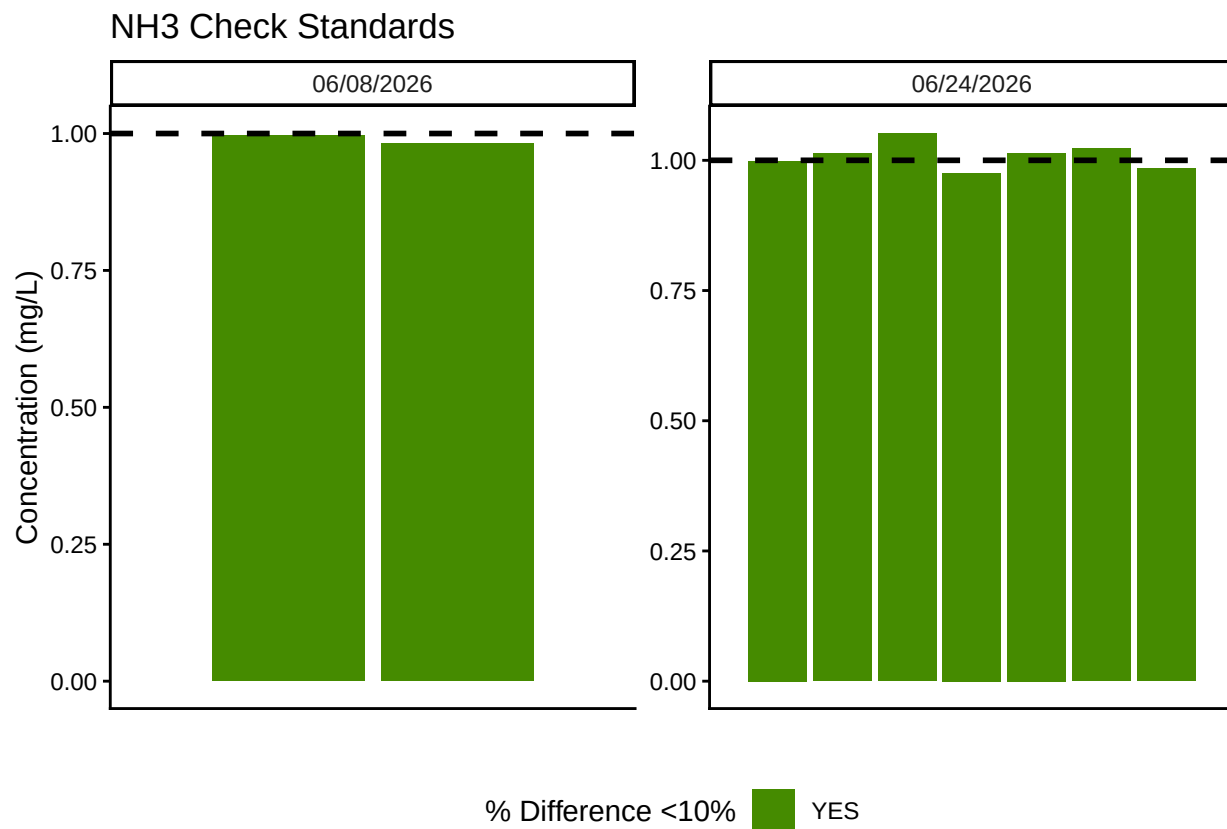
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	06/24/2026	Performance Check Within 25% - PROCEED	1.148001	1.034

0.5 Analyze the Check Standards

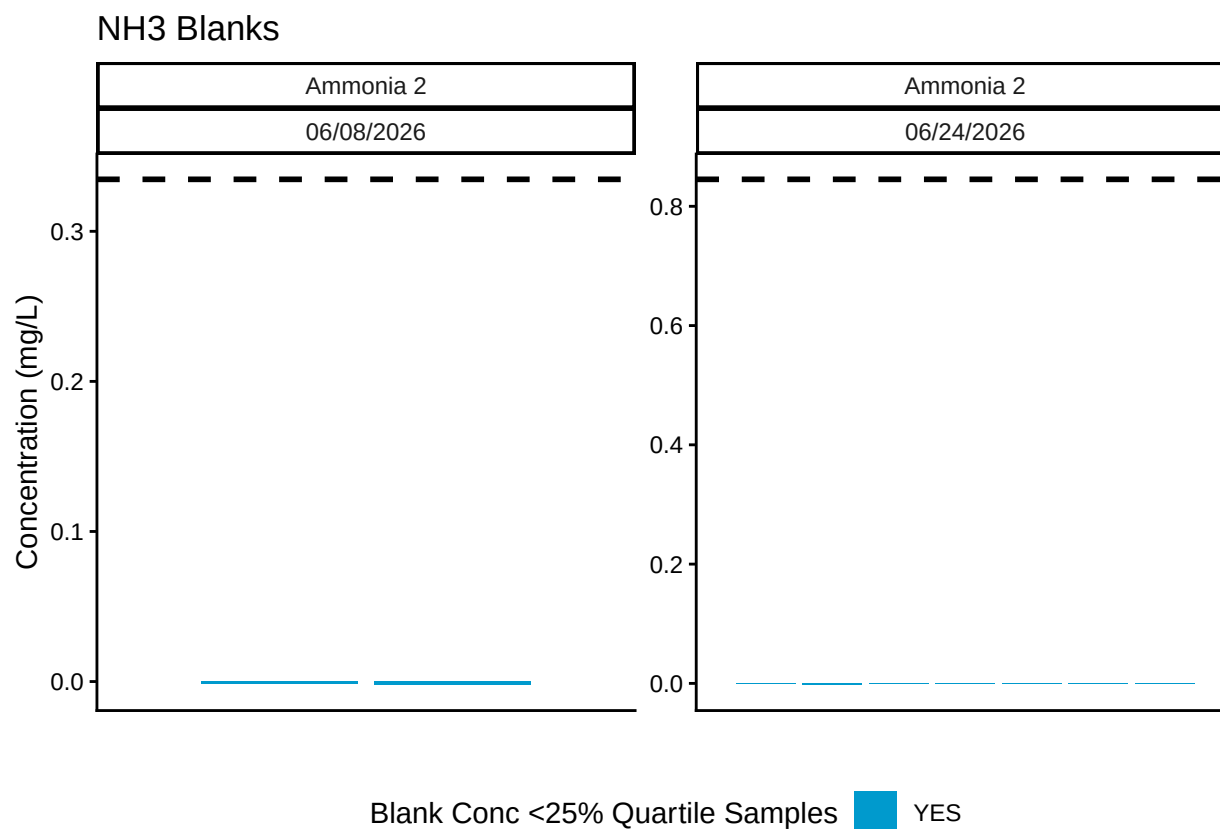
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	06/08/2026	RSV WITHIN RANGE - PROCEED	0.0236851	0.25
Ammonia 2	06/24/2026	RSV WITHIN RANGE - PROCEED	0.0236851	0.25



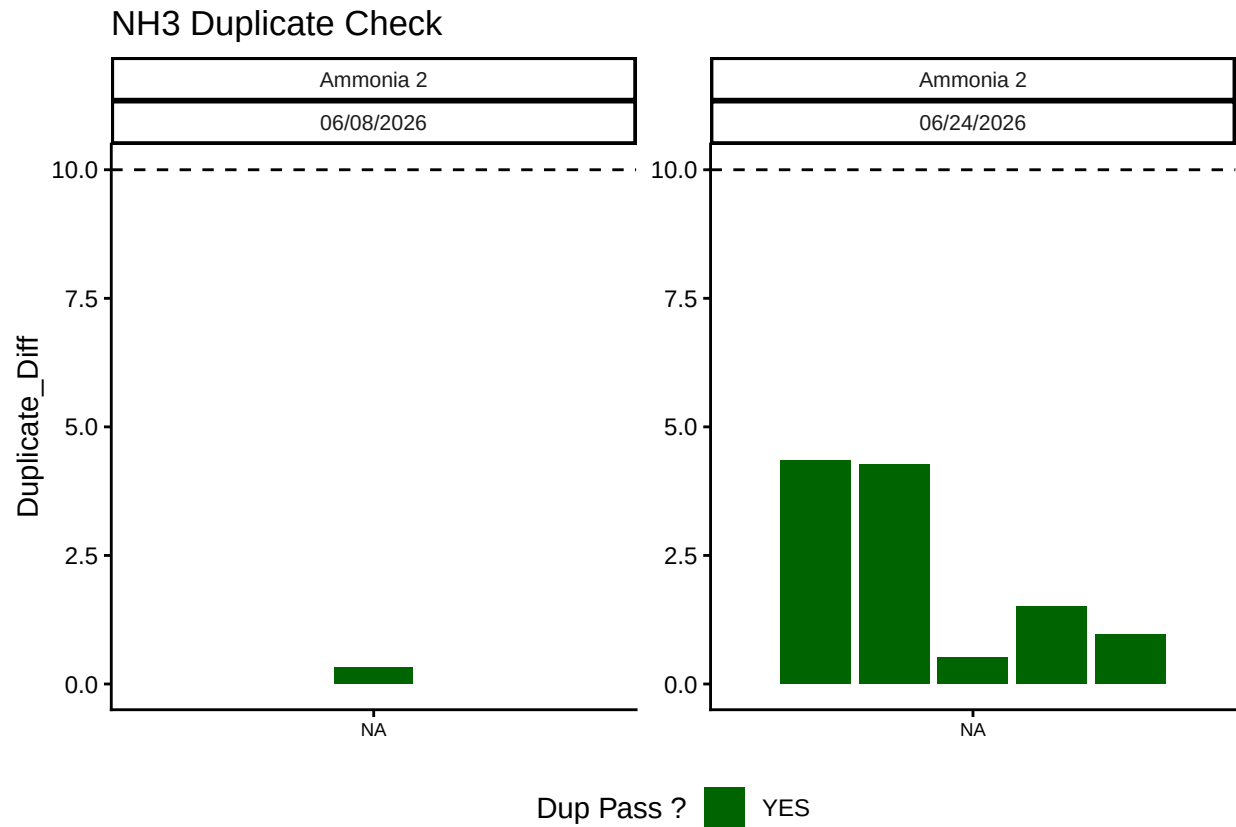
Test	Run_Date	CHK_Flag
Ammonia 2	06/08/2026	>60% of Checks Pass - PROCEED
Ammonia 2	06/24/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blkc_Conc	Quantile_25
Ammonia 2	06/08/2026	>60% of Blanks Pass - PROCEED	-0.002067	0.3346265
Ammonia 2	06/24/2026	>60% of Blanks Pass - PROCEED	-0.000556	0.8452105

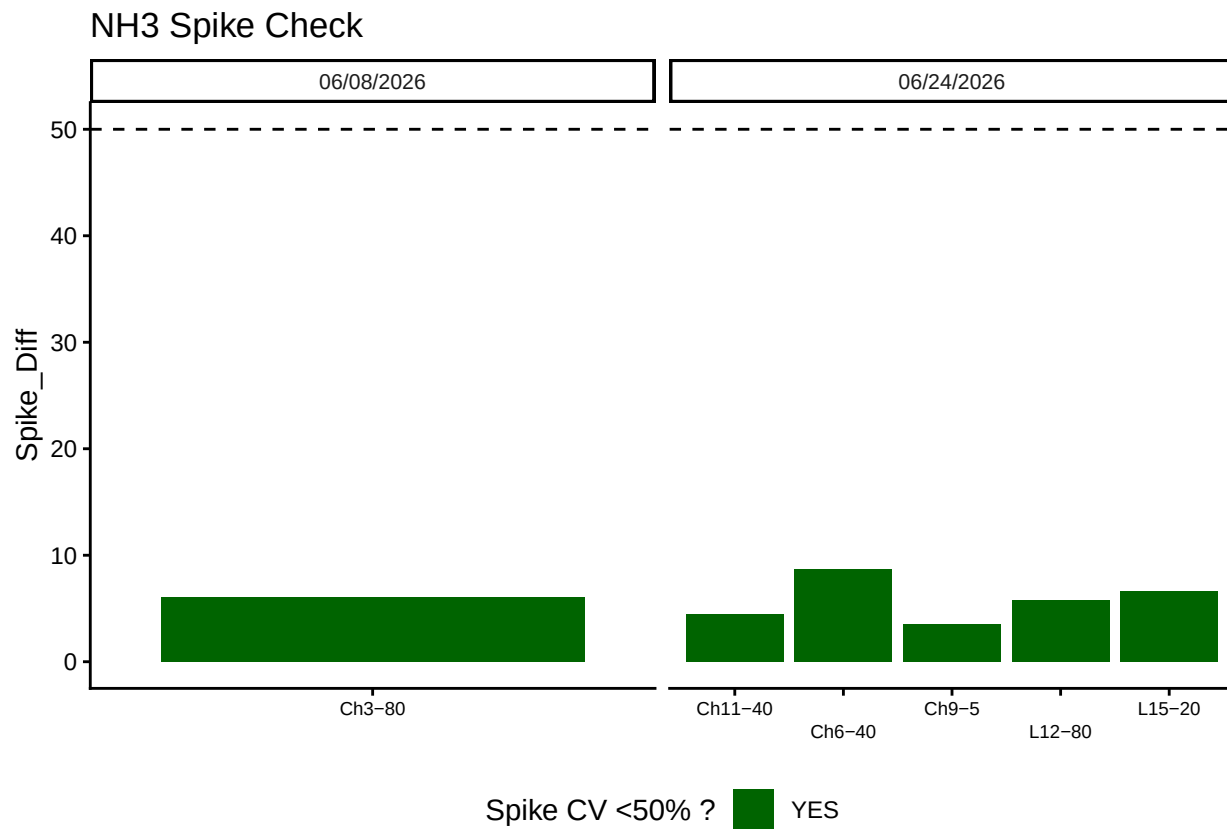


0.7 Analyze Duplicates



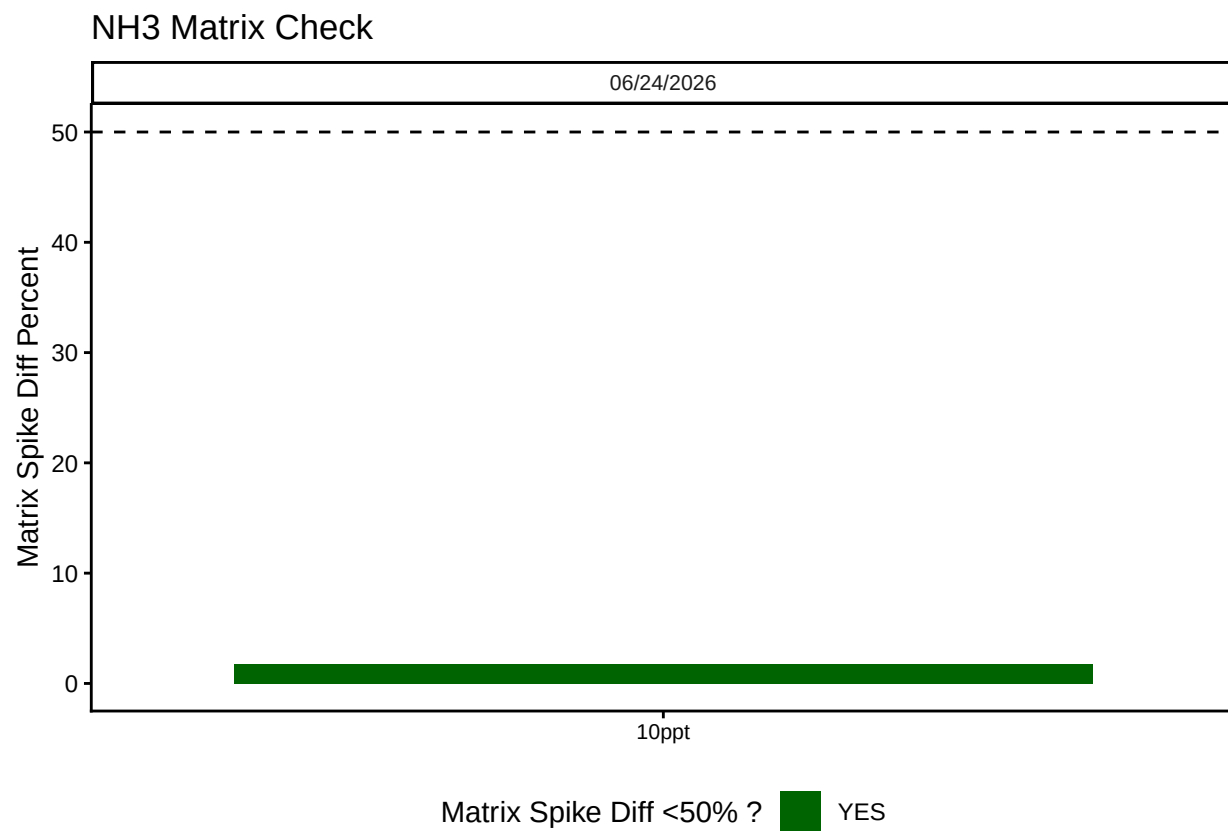
Test	Run_Date	Dup_Flags
Ammonia 2	06/08/2026	>60% of Dups Pass - PROCEED
Ammonia 2	06/24/2026	>60% of Dups Pass - PROCEED

0.8 Spikes



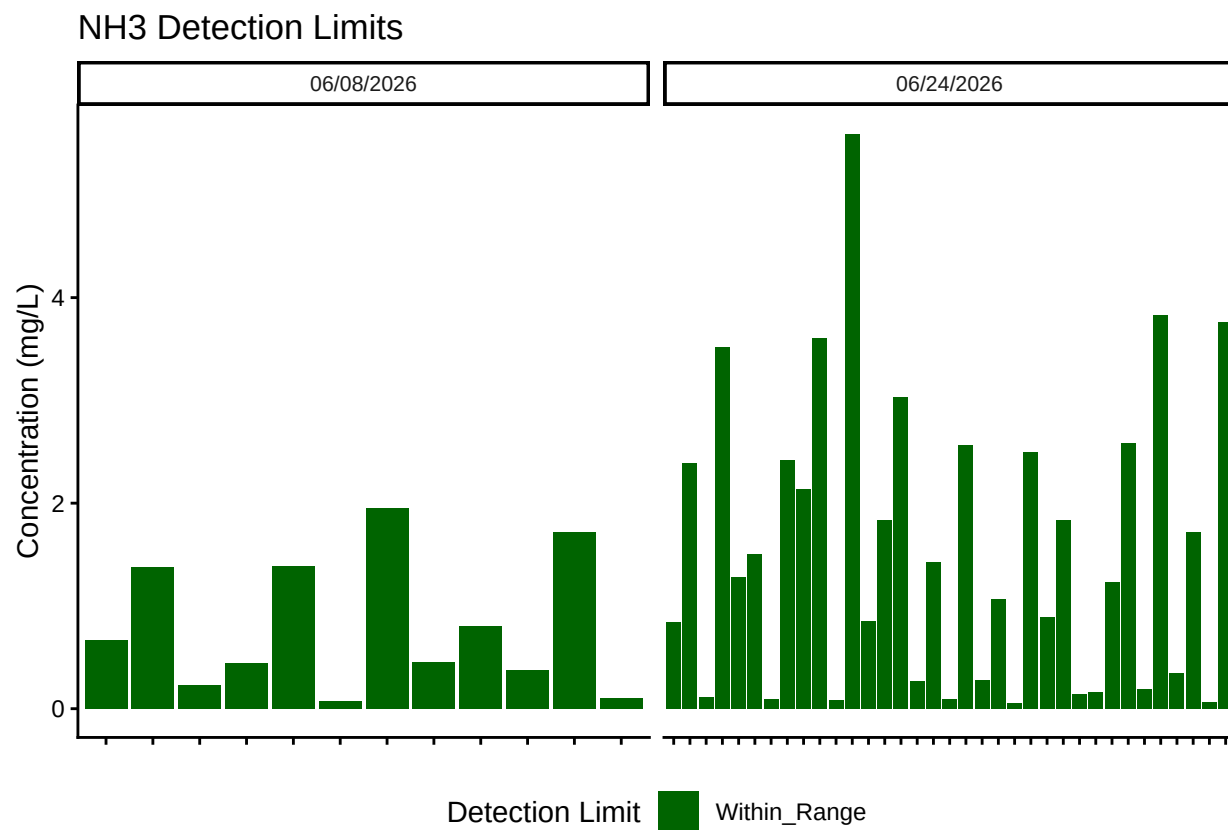
Test	Run_Date	Spike_Flags
Ammonia 2	06/08/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	06/24/2026	>60% of Spikes have a Diff <50% - PROCEED

0.9 Matrix Effects



##Unit Conversion

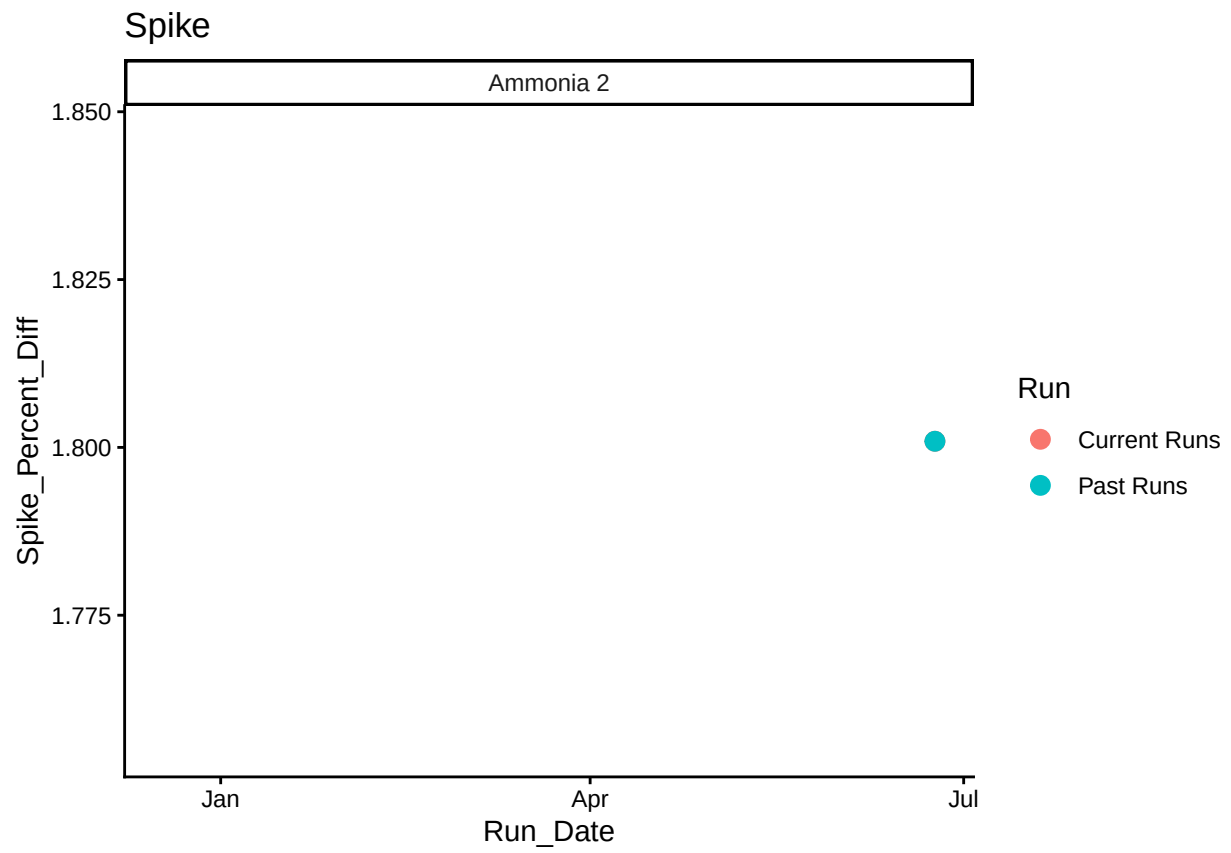
```
##Sample Flagging - Within range of standard curve
```

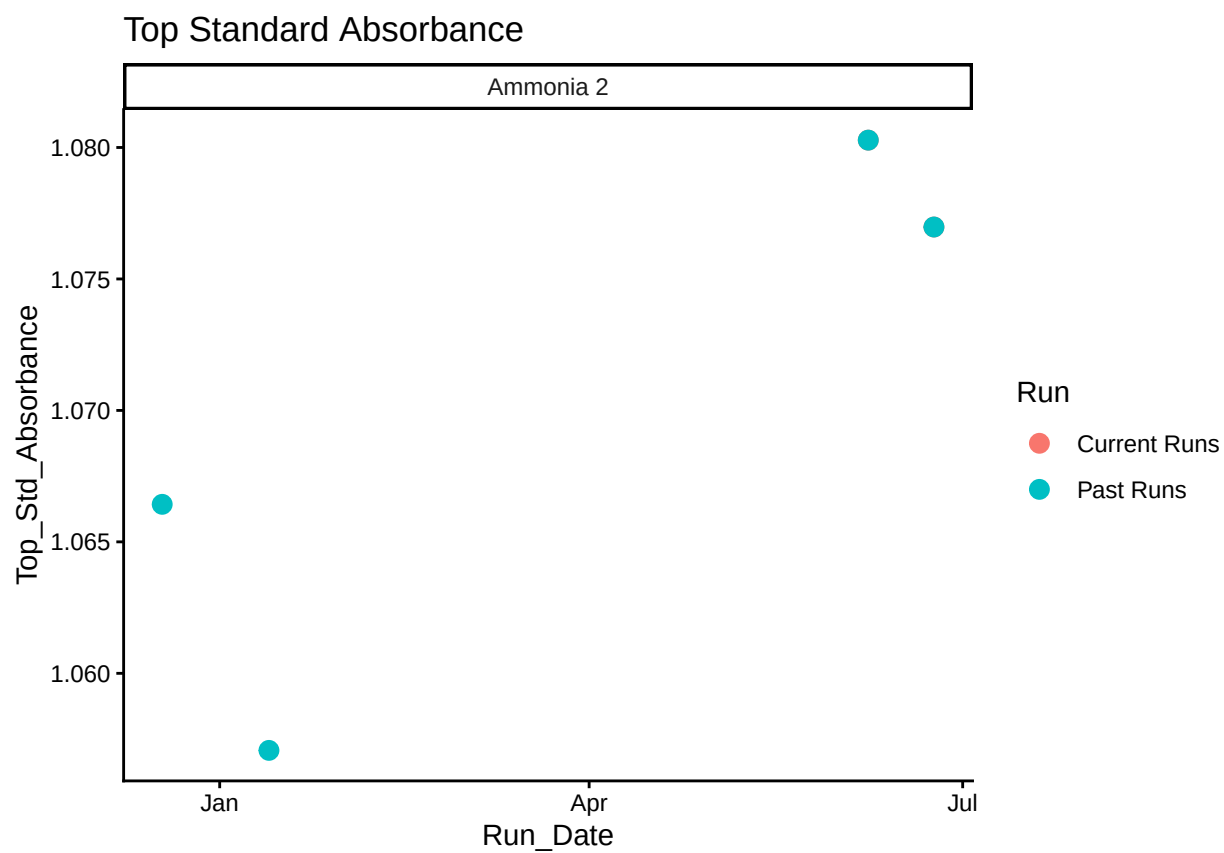


```
##Add QAQC Flags
```

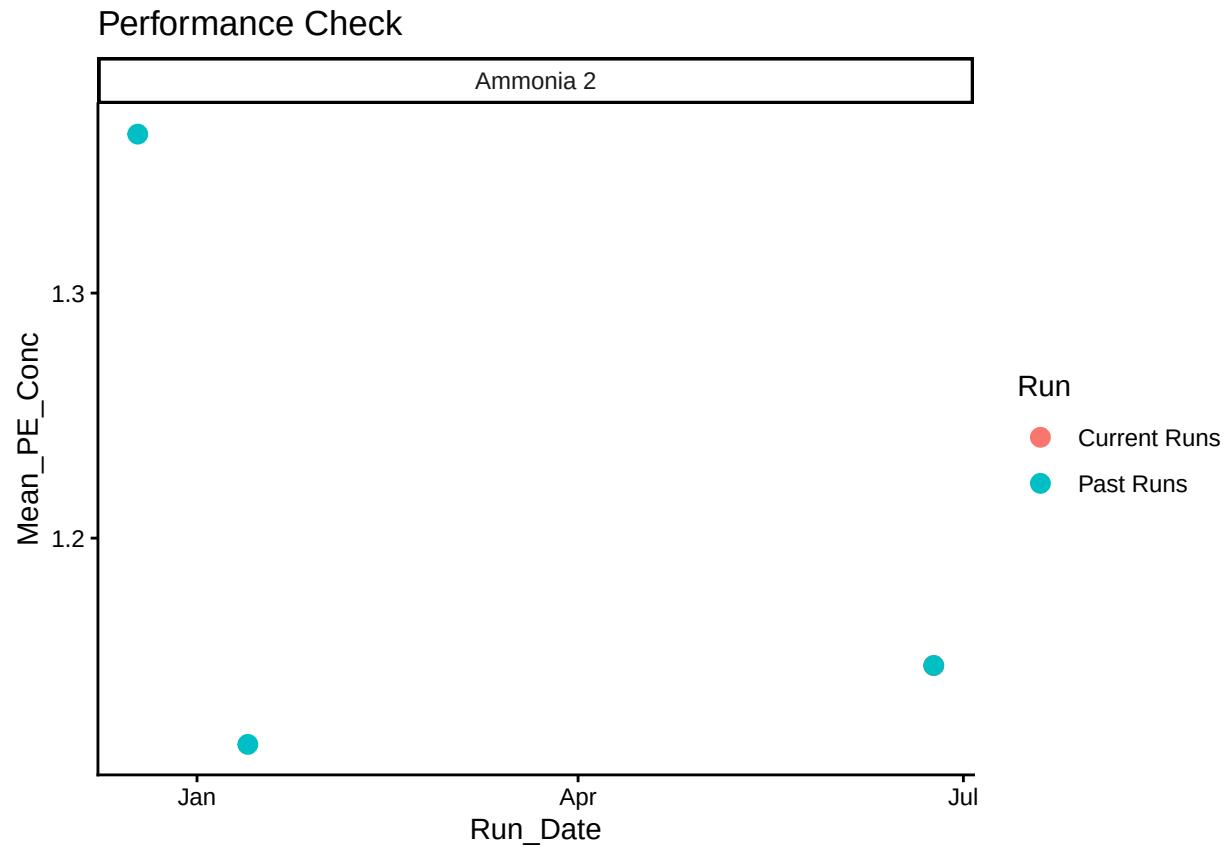
0.10 Create Log for QAQC

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_point()').
```





```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



0.11 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

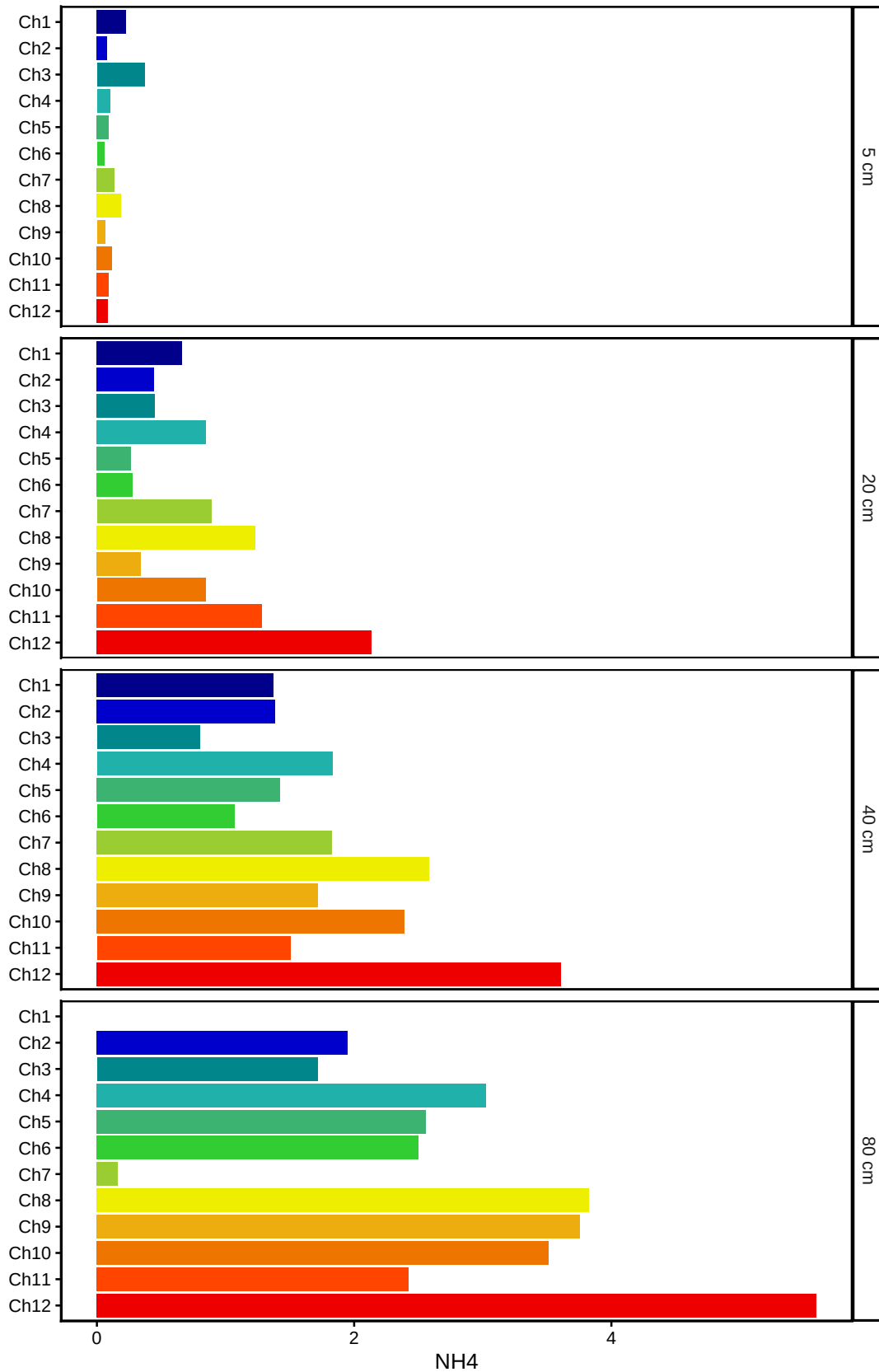
```
## [1] "Ch1-80"
```


0.12 Visualize Data

```
## Visualize Data
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## ('geom_col()').
```

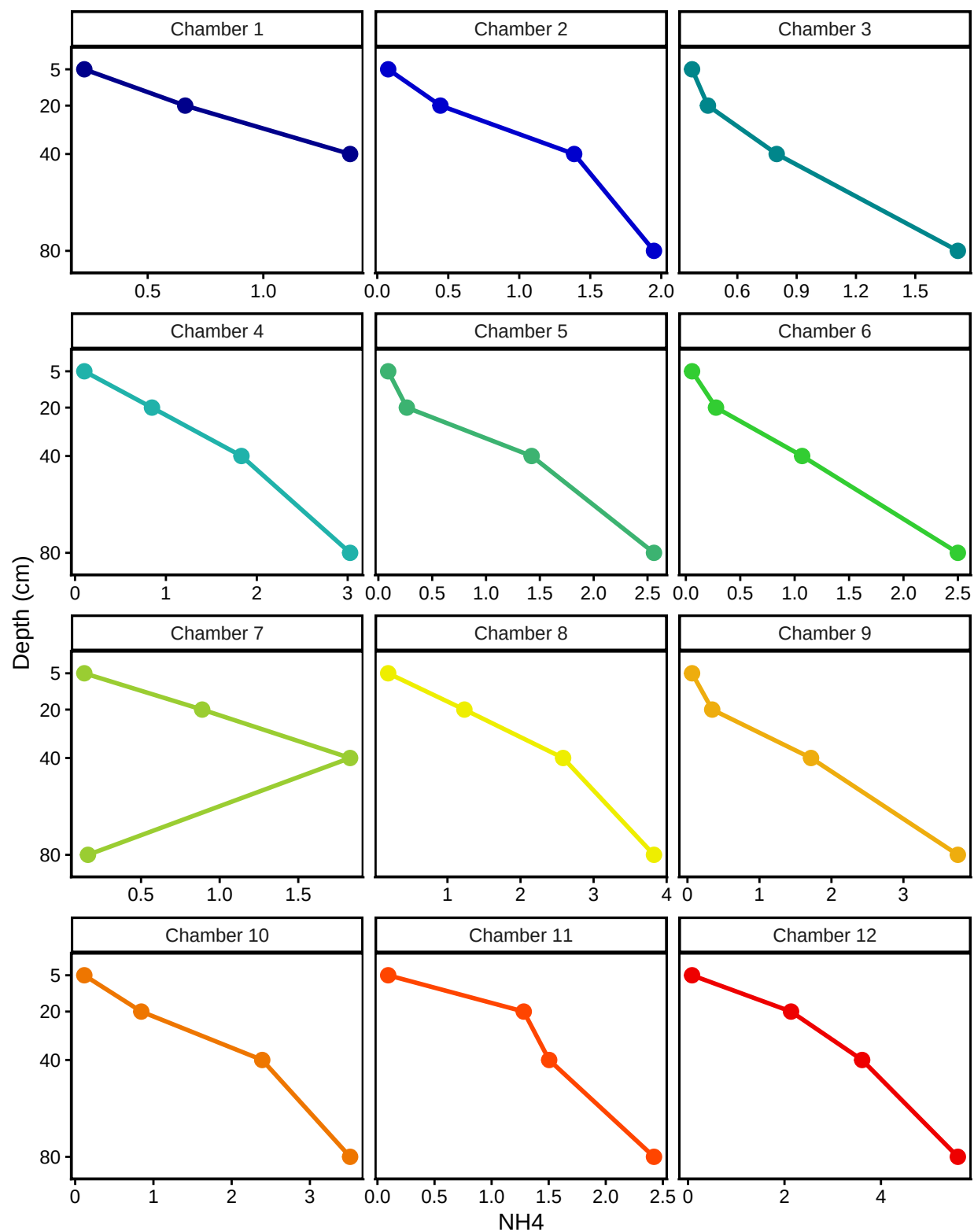
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```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_path()').
```

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0.13 Organize

#Export Data

#end